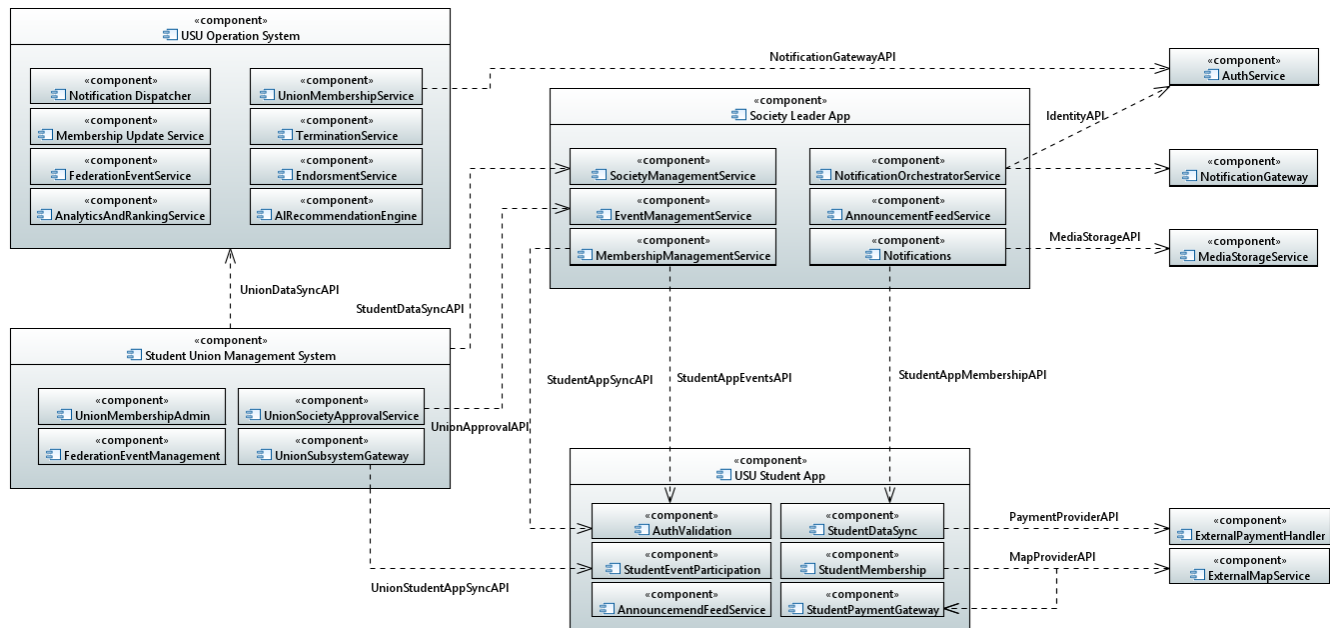


Task 4(b): Architectural Design of the USU System G29



The USU platform consists of four main subsystems—USU Operation System, Student Union Management System, Society Leader App, and USU Student App—plus shared external services for authentication, notifications, and third-party integrations. Each subsystem operates independently with its own backend services and communicates with others through well-defined APIs shown in the component diagram. This separation allows independent development and scaling while maintaining consistent integration across the platform.

The USU Operation System manages data across all participating universities through services like UnionMembershipService, FederationEventService, and AnalyticsAndRankingService. It receives society and event data from individual university unions, computes rankings and recommendations, and pushes this aggregated information back to the student-facing app. The Student Union Management System handles administration for each individual university including society approvals via UnionSocietyApprovalService, which is the approval step that precedes activation in FR-SL-1a. The Society Leader App enables leaders to activate societies, create events, and manage memberships through SocietyManagementService, EventManagementService, MembershipManagementService, and NotificationOrchestratorService. The USU Student App provides the student interface for discovering societies, joining memberships, and registering for events using StudentDataSync, StudentEventParticipation, and StudentMembership services.

The subsystems interact through several key APIs. UnionApprovalAPI allows the Society Leader App to check approval status from the Student Union Management System before activation. StudentAppSyncAPI, StudentAppEventsAPI, and StudentAppMembershipAPI propagate society, event, and membership data from the Society Leader App to the USU Student App so students can discover and join societies. UnionDataSyncAPI sends university-level data to the USU Operation System for cross-university analytics. NotificationGatewayAPI connects both the Society Leader App and USU Operation System to the shared NotificationGateway for email, SMS, and push delivery. IdentityAPI connects all subsystems to the shared AuthService for authentication and single sign-on.

This architecture supports the quality requirements defined for FR-SL-1a. Security is enforced through the shared AuthService and role-based access control in each subsystem, ensuring only authorised users can activate societies. Performance is improved by allowing the USU Student App to serve queries from its own local copy of society data rather than calling other subsystems on every search, and by using asynchronous APIs that provide immediate feedback during activation while processing slower tasks like notifications in the background. Reliability is enhanced through retry logic and idempotent API design so temporary failures in one subsystem do not block operations in another. Scalability is achieved by allowing each subsystem to scale independently and using asynchronous data propagation to handle peak loads during start-of-semester periods when many societies activate simultaneously.